

Tutorial Sheet

Assignment PS-0 due

2627-MA140

Some problems for MA140 tutorials in Week 2. These will not be graded, and will not contribute to your score for MA140.

Problem 1. (1 point)

Match the function to its domain:

NOTE: Enter the letter for the domain in the answer box. Letters may be used more than once.

___ 1. $f(x) = \frac{4}{x-5} + \frac{x+3}{x+4}$

___ 2. $f(x) = \frac{x-4}{x+3}$

___ 3. $f(x) = \frac{4}{x-5}$

___ 4. $f(x) = \frac{x-4}{x+4} + \frac{x+5}{x-4}$

- A. $\{x|x \text{ is a real number and } x \neq -4 \text{ and } x \neq 4\}$
- B. $\{x|x \text{ is a real number and } x \neq -3\}$
- C. $\{x|x \text{ is a real number and } x \neq 5\}$
- D. $\{x|x \text{ is a real number and } x \neq -4 \text{ and } x \neq 5\}$

Problem 2. (1 point)

If possible, find the indicated values for the rational function. If the value does not exist, enter *Does not exist*.

$$f(x) = \frac{3x^2 + x + 5}{x - 1}$$

a) $f(5) =$ _____

b) $f(0) =$ _____

c) $f(1) =$ _____

d) Find the domain of f : _____ Enter your answer using interval notation, e.g., If you think the answer is all of $\mathbb{R}$, you could enter this as $(-\infty, \infty)$ If it were $\mathbb{R}/10$, you could enter $(-\infty, 10) \cup (10, \infty)$ or $x < 10$ or $x > 10$

Problem 3. (1 point)

(a) On the same coordinate axes, sketch graphs (as accurately as possible) of the functions

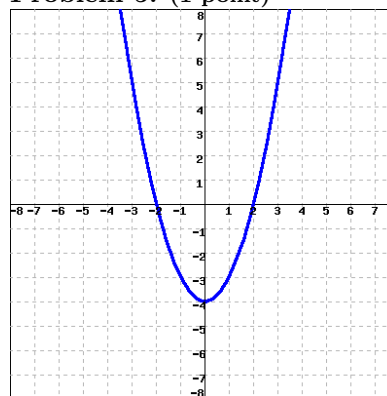
$$y = x^3 - 9x^2 + 2x + 3 \text{ and } y = -6x^2 + 20x + 3.$$

(b) Find the coordinates of all intersection points. (Enter your answer as a list of ordered pairs, e.g., $(1, 2)$, $(3, 4)$, ...)

Problem 4. (1 point)

Split into partial fractions:

$$\frac{12}{9 - x^2} = \frac{\quad}{\quad} + \frac{\quad}{\quad}$$

Problem 5. (1 point)

The graph of $f(x) = x^2 - 4$ is shown above.

a) Determine the x -intercepts of the graph. List the value(s) of x separated by a comma, or *none* if there are none.

$x =$ _____

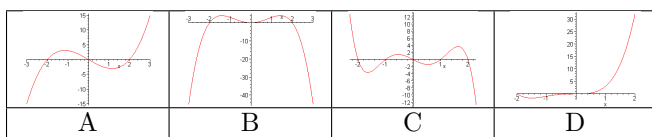
b) Find the solution(s) for: $x^2 - 4 = 0$. List the value(s) of x separated by a comma, or *none* if there are none.

Answer: _____

Problem 6. (1 point)

Match the functions with their graphs. Enter the letter of the graph below which corresponds to the function.
(Click on image for a larger view)

- ___ 1. $x^4 + 2x^3$
___ 2. $-x^5 + 5x^3 - 4x$
___ 3. $x(x^2 - 4)$
___ 4. $-x^2(x^2 - 4)$



Problem 7. (1 point)

Is the function below a rational function? If it is, write it in *reduced* form as a ratio of polynomials $\frac{p(x)}{q(x)}$. If it is not, enter *NONE* in both blanks. Do not leave any blanks empty. help (formulas)

$$\frac{x^2}{x-3} - \frac{7}{x-6} = \underline{\hspace{2cm}}$$

Problem 8. (1 point)

The partial fraction decomposition of $\frac{24}{(x-1)(x+1)}$ can be written in the form of $\frac{f(x)}{x-1} + \frac{g(x)}{x+1}$, where
 $f(x) = \underline{\hspace{2cm}}$,
 $g(x) = \underline{\hspace{2cm}}$.